1	2	3a PAT. CNTL #		4 TYPE OF BILL
		b. MED. REC. #	CLM003	1235
		5 FED. TAX NO.	6 STATEMENT COVERS PERIOD FROM	THROUGH 7
		221234567		

8 PATIENT NAME		9 PATIENT ADDRESS				
a Olivia		a 222 Redwood Ave				
b Brown		b Seattle	c WA	d 98109	e	

10 BIRTHDATE	11 SEX	12 ADMISSION DATE	13 HR	14 TYPE	15 SRC	16 DHR	17 STAT	18	19	20	21	22	23	24	25	26	27	28	29 ACDT STATE	30
07-18-82	F																			

	31 OCCURRENCE CODE	31 OCCURRENCE DATE	32 OCCURRENCE CODE	32 OCCURRENCE DATE	33 OCCURRENCE CODE	33 OCCURRENCE DATE	34 OCCURRENCE CODE	34 OCCURRENCE DATE	35 OCCURRENCE SPAN CODE	35 OCCURRENCE SPAN FROM	35 OCCURRENCE SPAN THROUGH	36 OCCURRENCE SPAN CODE	36 OCCURRENCE SPAN FROM	36 OCCURRENCE SPAN THROUGH	37	
a																a
b																b

38	39 VALUE CODES CODE	39 VALUE CODES AMOUNT	40 VALUE CODES CODE	40 VALUE CODES AMOUNT	41 VALUE CODES CODE	41 VALUE CODES AMOUNT
Olivia Brown 222 Redwood Ave, Seatte WA, 98109	a					
	b					
	c					
	d					

	42 REV. CD.	43 DESCRIPTION	44 HCPCS / RATE / HIPPS CODE	45 SERV. DATE	46 SERV. UNITS	47 TOTAL CHARGES	48 NON-COVERED CHARGES	49	
1	3283	NSCLC	88342		1	3000			1
2	7172	T3N2Mx	88360		1	7000			2
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23		***PAGE 1 OF 1***	***CREATION DATE***		***TOTALS***	10000			23

	50 PAYER NAME	51 HEALTH PLAN ID	52 REL. INFO	53 ASG. BEN.	54 PRIOR PAYMENTS	55 EST. AMOUNT DUE	56 NPI	
A		PL003					57	A
B							OTHER	B
C							PRV ID	C

	58 INSURED'S NAME	59 P REL	60 INSURED'S UNIQUE ID	61 GROUP NAME	62 INSURANCE GROUP NO.
A	Olivia Brown		P7890		
B					
C					

	63 TREATMENT AUTHORIZATION CODES	64 DOCUMENT CONTROL NUMBER	65 EMPLOYER NAME
A			
B			
C			

66 DX													68
	Z01.89	Z15.09	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	

69 ADMIT DX		70 PATIENT REASON DX	a	b	c	71 PPS CODE		72 ECI	a	b	c	73

74 PRINCIPAL PROCEDURE CODE	DATE	a. OTHER PROCEDURE CODE	DATE	b. OTHER PROCEDURE CODE	DATE	75
c. OTHER PROCEDURE CODE	DATE	d. OTHER PROCEDURE CODE	DATE	e. OTHER PROCEDURE CODE	DATE	

76 ATTENDING	NPI	QUAL	
LAST		FIRST	
77 OPERATING	NPI	QUAL	
LAST		FIRST	
78 OTHER	NPI	QUAL	
LAST		FIRST	
79 OTHER	NPI	QUAL	
LAST		FIRST	

80 REMARKS	81CC		
Refer to the attached	a		
report	b		
	c		
	d		

UB-04 NOTICE: THE SUBMITTER OF THIS FORM UNDERSTANDS THAT MISREPRESENTATION OR FALSIFICATION OF ESSENTIAL INFORMATION AS REQUESTED BY THIS FORM, MAY SERVE AS THE BASIS FOR CIVIL MONETARTY PENALTIES AND ASSESSMENTS AND MAY UPON CONVICTION INCLUDE FINES AND/OR IMPRISONMENT UNDER FEDERAL AND/OR STATE LAW(S).

Submission of this claim constitutes certification that the billing information as shown on the face hereof is true, accurate and complete. That the submitter did not knowingly or recklessly disregard or misrepresent or conceal material facts. The following certifications or verifications apply where pertinent to this Bill:

1. If third party benefits are indicated, the appropriate assignments by the insured /beneficiary and signature of the patient or parent or a legal guardian covering authorization to release information are on file. Determinations as to the release of medical and financial information should be guided by the patient or the patient's legal representative.
2. If patient occupied a private room or required private nursing for medical necessity, any required certifications are on file.
3. Physician's certifications and re-certifications, if required by contract or Federal regulations, are on file.
4. For Religious Non-Medical facilities, verifications and if necessary re-certifications of the patient's need for services are on file.
5. Signature of patient or his representative on certifications, authorization to release information, and payment request, as required by Federal Law and Regulations (42 USC 1935f, 42 CFR 424.36, 10 USC 1071 through 1086, 32 CFR 199) and any other applicable contract regulations, is on file.
6. The provider of care submitter acknowledges that the bill is in conformance with the Civil Rights Act of 1964 as amended. Records adequately describing services will be maintained and necessary information will be furnished to such governmental agencies as required by applicable law.
7. For Medicare Purposes: If the patient has indicated that other health insurance or a state medical assistance agency will pay part of his/her medical expenses and he/she wants information about his/her claim released to them upon request, necessary authorization is on file. The patient's signature on the provider's request to bill Medicare medical and non-medical information, including employment status, and whether the person has employer group health insurance which is responsible to pay for the services for which this Medicare claim is made.
8. For Medicaid purposes: The submitter understands that because payment and satisfaction of this claim will be from Federal and State funds, any false statements, documents, or concealment of a material fact are subject to prosecution under applicable Federal or State Laws.
9. For TRICARE Purposes:
   (a) The information on the face of this claim is true, accurate and complete to the best of the submitter's knowledge and belief, and services were medically necessary and appropriate for the health of the patient;
   (b) The patient has represented that by a reported residential address outside a military medical treatment facility catchment area he or she does not live within the catchment area of a U.S. military medical treatment facility, or if the patient resides within a catchment area of such a facility, a copy of Non-Availability Statement (DD Form 1251) is on file, or the physician has certified to a medical emergency in any instance where a copy of a Non-Availability Statement is not on file;
   (c) The patient or the patient's parent or guardian has responded directly to the provider's request to identify all health insurance coverage, and that all such coverage is identified on the face of the claim except that coverage which is exclusively supplemental payments to TRICARE-determined benefits;
   (d) The amount billed to TRICARE has been billed after all such coverage have been billed and paid excluding Medicaid, and the amount billed to TRICARE is that remaining claimed against TRICARE benefits;
   (e) The beneficiary's cost share has not been waived by consent or failure to exercise generally accepted billing and collection efforts; and,
   (f) Any hospital-based physician under contract, the cost of whose services are allocated in the charges included in this bill, is not an employee or member of the Uniformed Services. For purposes of this certification, an employee of the Uniformed Services is an employee, appointed in civil service (refer to 5 USC 2105), including part-time or intermittent employees, but excluding contract surgeons or other personal service contracts. Similarly, member of the Uniformed Services does not apply to reserve members of the Uniformed Services not on active duty.
   (g) Based on 42 United States Code 1395cc(a)(1)(j) all providers participating in Medicare must also participate in TRICARE for inpatient hospital services provided pursuant to admissions to hospitals occurring on or after January 1, 1987; and
   (h) If TRICARE benefits are to be paid in a participating status, the submitter of this claim agrees to submit this claim to the appropriate TRICARE claims processor. The provider of care submitter also agrees to accept the TRICARE determined reasonable charge as the total charge for the medical services or supplies listed on the claim form. The provider of care will accept the TRICARE-determined reasonable charge even if it is less than the billed amount, and also agrees to accept the amount paid by TRICARE combined with the cost-share amount and deductible amount, if any, paid by or on behalf of the patient as full payment for the listed medical services or supplies. The provider of care submitter will not attempt to collect from the patient (or his or her parent or guardian) amounts over the TRICARE determined reasonable charge. TRICARE will make any benefits payable directly to the provider of care, if the provider of care is a participating provider.

Medical Report

Patient Information:

- Member Id: P7890
- Name: Olivia Brown
- Address: 222 Redwood Ave,Seattle,WA, 98109

Date of Report: 08-01-2025

Referring Physician:

- Physician ID: PHY003
- Referring Physician: Dr. Sarah Lee

Clinical Summary:

A comprehensive genomic analysis and pathological examination were conducted on a lung mass with the following findings:

- **Cancer Type:** Non-small cell lung cancer (NSCLC)
- **Cancer Subtype:** Lung adenocarcinoma, poorly differentiated adenocarcinoma
- **Tumor Size:** 2.5 cm
- **Pathology Stage:** T3N2Mx, indicating the following:
    - **T3:** The tumor is locally advanced but has not invaded major structures like the chest wall or mediastinum.
    - **N2:** Regional lymph node involvement, with possible involvement of mediastinal or subcarinal lymph nodes.
    - **Mx:** Distant metastasis cannot be assessed at this time.

Genomic Biomarkers:

A detailed molecular analysis has been performed using Fluorescence in situ hybridization (FISH) to identify key mutations and genetic alterations in the tumor:

- **KRAS mutation:** Mutation detected in codon 12/13, which is commonly associated with resistance to EGFR-targeted therapies and indicates a more aggressive clinical course in lung adenocarcinomas.
- **EGFR Exon 19 Mutation:** Positive for an Exon 19 deletion, a well-known genetic alteration in lung adenocarcinoma that is associated with sensitivity to EGFR tyrosine kinase inhibitors (TKIs), such as osimertinib, erlotinib, and gefitinib.
- **EGFR Exon 21 Mutation:** Positive for an Exon 21 mutation (L858R), which is also a common activating mutation in EGFR, conferring sensitivity to EGFR TKIs.
- **EGFR Amplification:** Positive for EGFR gene amplification, which suggests an increased copy number of the EGFR gene. This may indicate the potential for therapeutic intervention with EGFR-targeted agents, even in cases where mutations are not present.
- **EML4/ALK Rearrangement:** Positive for the EML4/ALK fusion, which is a driver mutation found in a subset of NSCLC, particularly in non-smokers or light smokers. This alteration makes the tumor susceptible to ALK inhibitors such as crizotinib, ceritinib, or alectinib.

Conclusion:

The patient has been diagnosed with non-small cell lung cancer (NSCLC), specifically lung adenocarcinoma, with poorly differentiated features. The tumor measures 2.5 cm in size, and the pathology stage is classified as T3N2Mx, indicating local invasion and regional lymph node involvement, but with an unknown status regarding distant metastasis.
The tumor has several significant genomic biomarkers, including mutations and alterations that can guide treatment

Strategies:

- **KRAS codon 12/13 mutation:** Suggests resistance to EGFR-targeted therapy and could inform treatment planning, favoring non-EGFR-based therapies.
- **EGFR Exon 19 and Exon 21 mutations:** Indicate potential benefit from EGFR-targeted therapies, particularly tyrosine kinase inhibitors.
- **EGFR Amplification:** Further supports the potential use of EGFR-targeted therapies.
- **EML4/ALK Fusion:** The presence of this fusion suggests that the patient may benefit from ALK inhibitors as a treatment option.

The combination of EGFR and EML4/ALK alterations is unusual, and clinical decisions will need to account for both pathways in therapy selection.

Recommendations:

- **Molecularly targeted therapy:** Given the presence of EGFR mutations and EGFR amplification, EGFR tyrosine kinase inhibitors (TKIs) such as osimertinib or erlotinib should be considered as the first line of therapy. Additionally, the presence of EML4/ALK fusion makes ALK inhibitors (e.g., crizotinib) a viable treatment option, though treatment should focus on the most clinically appropriate mutation.
- **Staging and imaging:** A thorough imaging workup (CT, PET) is necessary to assess the extent of regional and distant metastasis, given the T3N2 stage.
- **Multidisciplinary approach:** A multidisciplinary team should review the treatment options, balancing the potential effectiveness of targeted therapies and traditional treatments, including chemotherapy or radiation.
- **Follow-up:** Continuous monitoring for treatment efficacy, disease progression, and potential resistance mechanisms is critical.

Report Prepared By:
PHY003
Dr. Sarah Lee